1. Ideation and Problem Statement

Project title

Electric Vehicle Market Performance & Insights

Background

Electric vehicles are represented in the supplied dataset across vehicle categories, manufacturers, markets, battery types, technical specifications, prices, and user-oriented measures. With 100,000 records and 37 fields, side-by-side inspection is difficult without structured summaries and interactive visualizations.

Problem statement

The project needs to turn a wide EV dataset into a clear, navigable analytical experience. Users should be able to compare the recorded vehicle specifications and commercial indicators, inspect distributions and relationships, and understand the limitations of the data.

Proposed solution

Use data profiling and descriptive analysis to prepare the data for Tableau. Build a dashboard for overview and comparison, a story for guided interpretation, and a website that introduces the project and embeds both published views.

Objectives

1. Verify the structure and quality of the supplied CSV.
2. Describe records by model year, manufacturer, country, region, category, and battery type.
3. Compare relevant specifications such as battery capacity, range, efficiency, motor power, charging attributes, price, and ratings.
4. Define every KPI and calculated field transparently.
5. Present findings in Tableau and a responsive web page.
6. Document limitations so that dataset-level patterns are not misrepresented as market-wide facts.

Primary audience

Students, instructors, EV-curious readers, and analysts who need a guided overview of vehicle attributes in the provided file.

Scope

Included: dataset audit, preparation documentation, descriptive analysis, Tableau dashboard/story, HTML website integration, quality checks, report, and demo.

Not established by the data alone: live market information, verified global market share, causal explanations, actual consumer behavior, real-world emissions savings, or forecasting. These would require provenance and methodology evidence not supplied with the CSV.

Success criteria

- Data counts reconcile with the supplied file.
- Missing values and unusual ranges are documented.
- Visualizations have clear titles, units, and aggregation definitions.
- Dashboard/story embed correctly and direct links remain available.
- Conclusions are traceable to calculations and include caveats.

Initial analytical questions

- How are records distributed across the 15 vehicle categories and 7 market regions?
- How do price and range vary across categories?
- How are battery capacity, range, and efficiency associated in this file?
- What is the distribution of annual sales values, and how sensitive are summaries to extreme values?
- How complete are the ratings, cycle-life, resale-value, and annual-sales fields?